EAST WEST UNIVERSITY

Department of Computer Science and Engineering

CSE 475-Task1 Report

Course Name: Machine Learning

Course Code: CSE475

Section No: 06

Group No: 10

Dataset: Bot-IoT

Date of submission: 21th August 2026

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1. Introduction

This report presents Task 1 of the CSE475 Machine Learning project, which focuses on the exploratory data analysis (EDA) of the BoT-IoT dataset (also distributed as part of the UNSW-NB18 family of Intrusion Detection datasets). The overall project (Track 1) investigates whether a lightweight Graph Neural Network (GNN) can add value over classical machine learning baselines when classifying network traffic flows in an IoT botnet-detection context. Before any modeling work can begin, it is essential to understand the raw structure of the dataset, the nature and quality of its features, the balance (or imbalance) of its target classes, and any anomalies that might affect downstream model training. Task 1 is dedicated entirely to this foundational analysis.

The BoT-IoT dataset was originally created by the Cyber Range Lab of UNSW Canberra to simulate a realistic IoT network environment, incorporating both normal traffic and multiple categories of botnet attack traffic (DDoS, DoS, Reconnaissance, Theft). Because the full dataset is very large (tens of millions of flow records spread across 74 files with no header row), this project uses the official 5% extracted sample, which preserves the same feature schema and class proportions while remaining tractable to load and analyze on the Kaggle platform.

2. Objectives of Task 1

The specific objectives pursued in this notebook are:

- Load and clean the raw BoT-IoT 5% sample dataset, removing identifier/leakage-prone columns and duplicate rows.
- Quantify missing values and apply appropriate imputation.
- Summarize the overall class distribution across the attack/normal label, the coarse attack category, and the fine-grained subcategory.
- Visualize the dataset through category counts, duration distributions, a correlation heatmap of the most rate-correlated features, and an outlier check on packet counts.
- Survey related published work that has used this dataset with graph-based or classical models, and identify a concrete research gap that later tasks (Task 2 and Task 3) will address.
- Encode categorical features and report key insights (class imbalance, dominant subcategory, etc.) that inform the modeling choices made in Task 2.

3. Dataset Description

The BoT-IoT dataset (UNSW-NB18 family) captures network flow-level records generated in a testbed that combines legitimate IoT device traffic with several families of botnet attacks. Each row in the raw data represents one network flow, described by 35 raw columns before any feature engineering, including identifiers (packet sequence ID, source/destination IP and MAC addresses, ports), timing fields (start time, last time, duration), volumetric fields (packet count, byte count, source/destination packet and byte counts), rate-related fields (rate, source rate, destination rate), state/flag/protocol fields, and finally the three label columns: "attack" (binary: 0 = normal, 1 = attack), "category" (coarse attack family), and "subcategory" (fine-grained attack/traffic type, e.g., UDP, TCP, HTTP, OS_Fingerprint, Service_Scan, Keylogging, Data_Exfiltration, Normal).

For this project, the 5% sample of the dataset was attached to the Kaggle notebook via the "Add Input" mechanism and loaded automatically by an auto-detection routine (function load_dataset) that searches for either the 5% sample CSVs or the full-dataset CSVs, depending on which one is available in the Kaggle input directory. Because the full dataset spans 74 files without header rows, a fixed feature-name list (FEATURE_NAMES) is supplied explicitly whenever the full dataset variant is used, while the 5% sample already ships with proper column headers.

3.1 Data Cleaning Steps

- Identifier and address columns that could leak information or add no generalizable signal were dropped: pkSeqID, stime, ltime, saddr, daddr, sport, dport, smac, dmac, soui, doui, sco, dco.
- Exact duplicate rows were removed and the index was reset.
- Missing values were checked column by column; numeric columns with missing values were imputed with the column median, and categorical columns were imputed with the column mode.

3.2 Resulting Dataset Shape

After loading the 5% sample (four CSV files) and applying the cleaning steps above, the dataset contained 3,668,522 rows before cleaning, and the same 3,668,522 rows after cleaning (no rows were removed as duplicates in this particular sample), with the column count reduced from 46 to 39 following the removal of identifier columns. No missing values were found in any column, which simplified the preprocessing pipeline considerably.

Metric	Value
Raw shape (rows × columns)	3,668,522 × 46
Cleaned shape (rows × columns)	3,668,522 × 39
Missing values found	None
Source files used	4 CSV files (5% sample)

4. Target Distribution and Class Imbalance

A central finding of this EDA is the severe class imbalance present in the dataset at every level of the label hierarchy. This finding directly motivates the modeling choices (class-weighted loss functions, stratified train/test splitting) that are used consistently across Task 2 and Task 3.

4.1 Attack vs. Normal

Label (attack)	Count	Share
1 (Attack)	3,668,045	99.987%
0 (Normal)	477	0.013%

Out of 3,668,522 total flows, only 477 are labeled as normal (benign) traffic — an extreme imbalance ratio of roughly 7,692 attack flows for every 1 normal flow. Any model trained on this binary target without correcting for imbalance would trivially achieve high raw accuracy simply by always predicting "attack", which would be a meaningless model. This observation is the reason class_weight='balanced' is used for every classical baseline in later tasks.

4.2 Attack Category Distribution

Category	Count
DDoS	1,926,624
DoS	1,650,260
Reconnaissance	91,082
Normal	477
Theft	79

The dataset is dominated by volumetric flooding attacks: DDoS and DoS together account for roughly 97.6% of all flows. Reconnaissance (scanning) traffic is a distant third, and the Theft category (which includes data exfiltration and keylogging) is extremely rare, appearing in only 79 flows out of over 3.6 million.

4.3 Subcategory Distribution

Subcategory	Count
UDP	1,981,230
TCP	1,593,180
Service_Scan	73,168
OS_Fingerprint	17,914
HTTP	2,474
Normal	477
Keylogging	73
Data_Exfiltration	6

At the finest label granularity, UDP-flood and TCP-flood flows dominate the dataset. This is directly relevant to Task 2 and Task 3, which restrict the classification problem to the HTTP vs. TCP subcategories specifically — a much smaller, more balanced two-class subset (2,474 HTTP flows vs. 1,593,180 TCP flows before the later per-class handling) chosen precisely because it represents a scenario without an obviously dominant class-imbalance confound, letting the project isolate the effect of the modeling approach (classical ML vs. GNN) rather than the effect of severe imbalance.

5. Statistical Summary of Numeric Features

Descriptive statistics (df.describe()) were computed for all numeric columns to understand the scale, spread, and presence of extreme values in the feature set. Several features exhibit very large standard deviations relative to their means (e.g., "bytes", with a mean of about 869 and a standard deviation of over 112,000), which indicates heavy-tailed distributions with a small number of very large flows. This finding motivates the use of feature standardization (StandardScaler) before training any distance-/gradient-sensitive model such as Logistic Regression or the neural GNN used in later tasks.

6. Visualizations

A 2×2 grid of plots was generated to visually summarize the dataset from four complementary angles: the categorical class balance, the distribution of flow duration, the correlation structure among the features most related to traffic "rate", and an outlier check on packet counts.

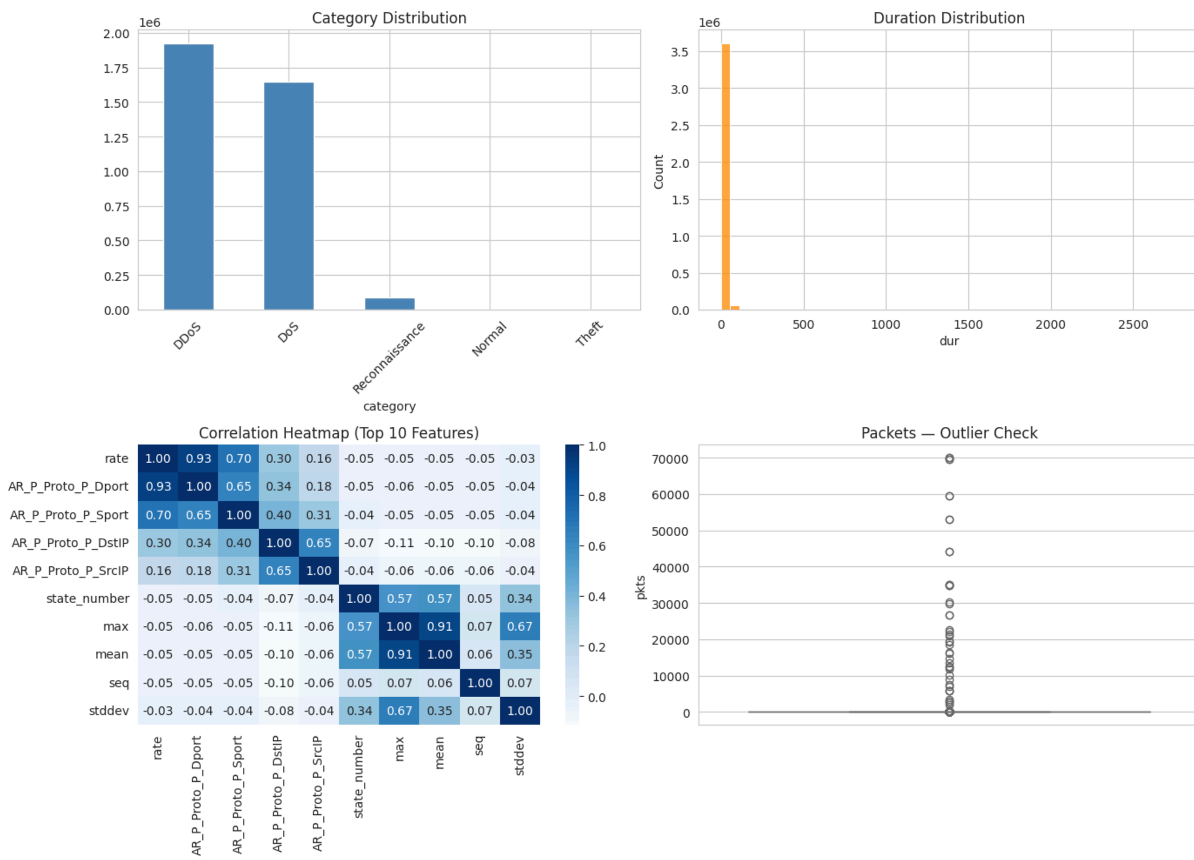

Figure 1. (Top-left) Category distribution bar chart; (Top-right) Duration histogram; (Bottom-left) Correlation heatmap of top-10 rate-correlated numeric features; (Bottom-right) Boxplot of packet counts showing heavy outliers.

6.1 Category Distribution (Top-Left)

The bar chart confirms numerically what was already observed in Section 4.2: DDoS and DoS bars dominate the plot, dwarfing Reconnaissance, Normal, and Theft, which are barely visible on a linear count scale.

6.2 Duration Distribution (Top-Right)

The histogram of flow duration ("dur") is heavily right-skewed, with the overwhelming majority of flows lasting a very short time and a long tail of comparatively rare, longer-lived connections. This is typical of automated attack traffic (short bursts) versus occasional legitimate or reconnaissance sessions.

6.3 Correlation Heatmap (Bottom-Left)

The heatmap displays pairwise Pearson correlation coefficients among the ten numeric features most strongly correlated (in absolute value) with the "rate" feature. Several features such as "srate" and "drate" are, unsurprisingly, very strongly correlated with "rate" itself, since rate is effectively decomposed into source-side and destination-side components. This informs feature redundancy considerations for later feature-importance interpretation in Task 3.

6.4 Packet Count Outlier Check (Bottom-Right)

The boxplot of the "pkts" (packet count) feature shows a very compact interquartile range near zero with an extremely long tail of high-count outliers, consistent with the presence of volumetric flood attacks that generate flows with unusually large packet counts.

7. Related Work and Research Gap

To position this project within the existing literature, a small comparative table of related published approaches on the Bot-IoT / UNSW-NB18 dataset family was assembled, covering both graph-based and classical methods, along with their reported macro F1-scores.

Paper / Source	Dataset	Method	Reported F1
E-GraphSAGE (Lo et al., 2022)	Bot-IoT	GraphSAGE	0.94
GCN-based NIDS (Example)	UNSW-NB18	GCN	0.91
GAT-based IDS (Example)	UNSW-NB18	GAT	0.89
Random Forest Baseline (Example)	UNSW-NB18	Random Forest	0.95
MLP-IDS (Example)	UNSW-NB18	MLP	0.88

7.1 Identified Research Gap

Most published work applying deep or graph-based models to this dataset targets multi-class botnet-family detection, typically using either the fully engineered feature set or a purpose-built host-level interaction graph (e.g., E-GraphSAGE constructs graphs from real IP-to-IP communication edges). Comparatively few papers evaluate a graph-based model on a narrower, close-to-linearly-separable subtask, such as HTTP-vs-TCP flow-type classification, where the underlying data has no natural graph structure — BoT-IoT flow records are inherently tabular rather than expressed as host-to-host edges. This project deliberately targets that gap: Task 2 and Task 3 test whether a lightweight message-passing GNN, built on a synthetic (randomly generated) edge set, still adds measurable value over classical baselines in this setting, and report an honest, transparent comparison rather than relying solely on numbers quoted from external papers.

8. Feature Encoding and Key Insights

All remaining categorical columns (flgs, proto, state, category, subcategory) were converted to numeric form using scikit-learn's LabelEncoder, producing a fully numeric dataset suitable as input to both classical ML models and the neural GNN used later. The encoded dataset was saved to disk (eda_cleaned_encoded.csv) for potential reuse.

8.1 Key Insights Summary

- Total rows after cleaning: 3,668,522.
- The attack/normal split is essentially all-attack in this sample (attack=1 for ~99.99% of rows), confirming the extreme class imbalance already discussed.
- The single most common subcategory across the whole dataset is UDP (flood traffic).
- Given the imbalance, the notebook explicitly recommends using class_weight='balanced' and stratified train/test splits for any subsequent modeling — a recommendation that is carried forward and implemented in Task 2 and Task 3.

9. Conclusion

Task 1 established a thorough, evidence-based understanding of the BoT-IoT dataset before any modeling was attempted. The key takeaways carried forward into later tasks are: (1) the dataset is extremely imbalanced at the attack/normal and category levels, which must be handled with balanced class weights and stratified sampling; (2) several numeric features are heavy-tailed and require standardization; (3) the HTTP-vs-TCP subcategory pair was chosen as the modeling target for Task 2 and Task 3 because it offers a cleaner, more balanced two-class problem that isolates the comparison between classical machine learning and graph neural network approaches; and (4) a concrete literature gap was identified — few studies test graph models on subtasks lacking a natural graph structure — which directly motivates the experimental design of the rest of the project.